

**CLUSTER FOR CLEAN ENERGY AND  
CLIMATE CHANGE (CCECC), BOWEN  
UNIVERSITY, IWO.**



**1<sup>ST</sup> NATIONAL CONFERENCE ON CLEAN  
ENERGY AND CLIMATE CHANGE  
15-16<sup>TH</sup> APRIL, 2026**

**C C E C C P R O C E E D I N G S**

**Advancing Clean Energy  
Solutions for Climate-  
Resilient and Sustainable  
Development in Nigeria**

## Climate modeling of flood frequency and intensity in Bayelsa State using SWAT+ for climate change adaptation

IJAOLA Opololaoluwa Oladimarun<sup>1</sup> and Okuroghoboye Diepreye ITUGHA<sup>2</sup>

<sup>12</sup>Department of Civil Engineering, Faculty of Engineering, Federal University Otuoke, Otuoke, Bayelsa State, Nigeria

opololaoluwaijaola121@gmail.com, [Ijaolao@fuotuke.edu.ng](mailto:Ijaolao@fuotuke.edu.ng)<sup>1</sup>.

[diepitu@yahoo.com](mailto:diepitu@yahoo.com)<sup>2</sup>

### Abstract

Climate change is increasingly altering hydrological processes in low-lying coastal environments, intensifying flood risks in vulnerable regions such as Bayelsa State in the Niger Delta, Nigeria. This study applied the Soil and Water Assessment Tool (SWAT+) to evaluate the impact of climate variability on flood frequency and intensity in the region. Hydrological modeling was conducted using a 30 m Shuttle Radar Topography Mission (SRTM) Digital Elevation Model, Landsat-derived land use data, FAO soil maps, and climate data obtained from the Nigerian Meteorological Agency (NiMet) and CORDEX-Africa datasets. The model was calibrated and validated using observed streamflow data through the SWAT-CUP SUFI-2 algorithm. Model performance indicated good agreement with observed data, with calibration statistics of NSE = 0.72,  $R^2 = 0.78$ , and PBIAS = -5.2%, and validation statistics of NSE = 0.65,  $R^2 = 0.70$ , and PBIAS = +8.3%. Historical analysis (2001–2024) revealed mean annual rainfall of approximately 3,055 mm, with a maximum flood peak of 45 m<sup>3</sup>/s recorded in 2009. Future projections under the RCP 8.5 scenario (2025–2045) indicate increased rainfall variability, rising temperatures (+1.9 °C), and a projected 15% increase in mean streamflow. Peak flood discharge is expected to increase significantly, suggesting heightened flood magnitude and frequency. These findings highlight the urgent need for climate-informed flood management strategies and resilient infrastructure planning in Bayelsa State.

### Keywords

Climate change, flood frequency, SWAT+, hydrological modeling, Bayelsa State, Niger Delta

## 1. Introduction

Climate change has become one of the most critical environmental challenges of the 21st century, with far-reaching impacts on global hydrological systems. Rising global temperatures, driven by increased greenhouse gas emissions, have intensified the hydrological cycle, resulting in changes in precipitation patterns, evapotranspiration rates, and the frequency of extreme weather events such as floods and droughts (Intergovernmental Panel on Climate Change, 2021). Among these impacts, flooding has emerged as one of the most devastating natural hazards, particularly in low-lying coastal and deltaic regions where physical and climatic conditions amplify vulnerability.

Globally, the increasing intensity and variability of rainfall have been linked to more frequent and severe flood events. Warmer atmospheric conditions enhance moisture-holding capacity, leading to

extreme precipitation and higher surface runoff (Trenberth, 2011). These changes significantly affect watershed hydrology by reducing infiltration and increasing peak discharge, thereby elevating flood risk (Milly et al., 2008). In developing countries, where infrastructure is often inadequate and adaptive capacity is limited, the consequences of such hydrological extremes are particularly severe, resulting in widespread socio-economic and environmental impacts.

Nigeria has experienced recurrent flooding in recent decades, with increasing severity attributed to climate variability and land use changes. The Niger Delta region, in particular, is highly vulnerable due to its low elevation, extensive river networks, and proximity to the Atlantic Ocean. Bayelsa State, located at the core of the Niger Delta, is one of the most flood-prone areas in the country. The region is characterized by annual rainfall ranging from 2,500 mm to 4,000 mm, extensive wetlands, and poorly drained soils, all of which contribute to frequent inundation (Ayanlade et al., 2006; Nwankwor et al., 2019). Seasonal flooding is often exacerbated by river overflow, tidal influences, and increasing rainfall variability associated with climate change, leading to significant damage to infrastructure, agriculture, and human settlements.

Understanding the dynamics of flood occurrence under changing climatic conditions is essential for effective water resources management and climate adaptation planning. Traditional methods of flood analysis, which rely on historical observations, are increasingly inadequate due to the non-stationary nature of climate systems (Milly et al., 2008). Consequently, there is a growing need for robust hydrological modeling tools that can simulate watershed processes and predict future hydrological responses under different climate scenarios.

The Soil and Water Assessment Tool (SWAT) are a physically based, semi-distributed hydrological model widely used for simulating the impact of land use, soil properties, and climate variability on watershed hydrology (Arnold et al., 1998). The enhanced version, SWAT+, provides improved representation of hydrological processes and flexible spatial discretization, making it suitable for complex environments such as the Niger Delta. SWAT+ has been successfully applied in various studies to assess climate change impacts on streamflow, runoff, and flood behavior (Neitsch et al., 2011; Moriasi et al., 2007). Its ability to integrate Geographic Information System (GIS) data with climate projections makes it particularly valuable for data-scarce regions.

Recent advances in climate modeling, including the use of Representative Concentration Pathways (RCPs), have enabled the simulation of future climate conditions under different greenhouse gas emission scenarios. The RCP 8.5 scenario, which represents a high-emission pathway, is often used to assess worst-case climate impacts, including increased temperature and precipitation variability (IPCC, 2021). Integrating such climate projections with hydrological models like SWAT+ allows for a comprehensive assessment of future flood risks.

Despite the increasing application of hydrological models in climate studies, there remains a gap in localized assessments of flood dynamics in the Niger Delta, particularly in Bayelsa State. Many previous studies have focused on broader regional analyses without detailed watershed-scale modeling. Given the region's high vulnerability, there is a need for detailed, location-specific studies that combine climate projections with hydrological modeling to evaluate future flood risks. Similarly there are still lack of detailed, watershed-scale modeling that integrates climate projections with hydrological processes to assess future flood dynamics in Bayelsa State. Hence this study addresses

this gap by applying the SWAT+ model to: simulate rainfall–runoff processes, analyze flood frequency and intensity, and assess future hydrological responses under climate change scenarios.

## 2. Materials and Methods

### 2.1 Study Area

The study area is Bayelsa State in southern Nigeria (Latitudes 4°15'N–5°23'N and Longitudes 5°22'E–6°45'E). The region is characterized by:

- Low elevation (< 50 m),
- Tropical humid climate,
- Extensive wetlands and floodplains,
- High rainfall (2,500–4,000 mm annually).

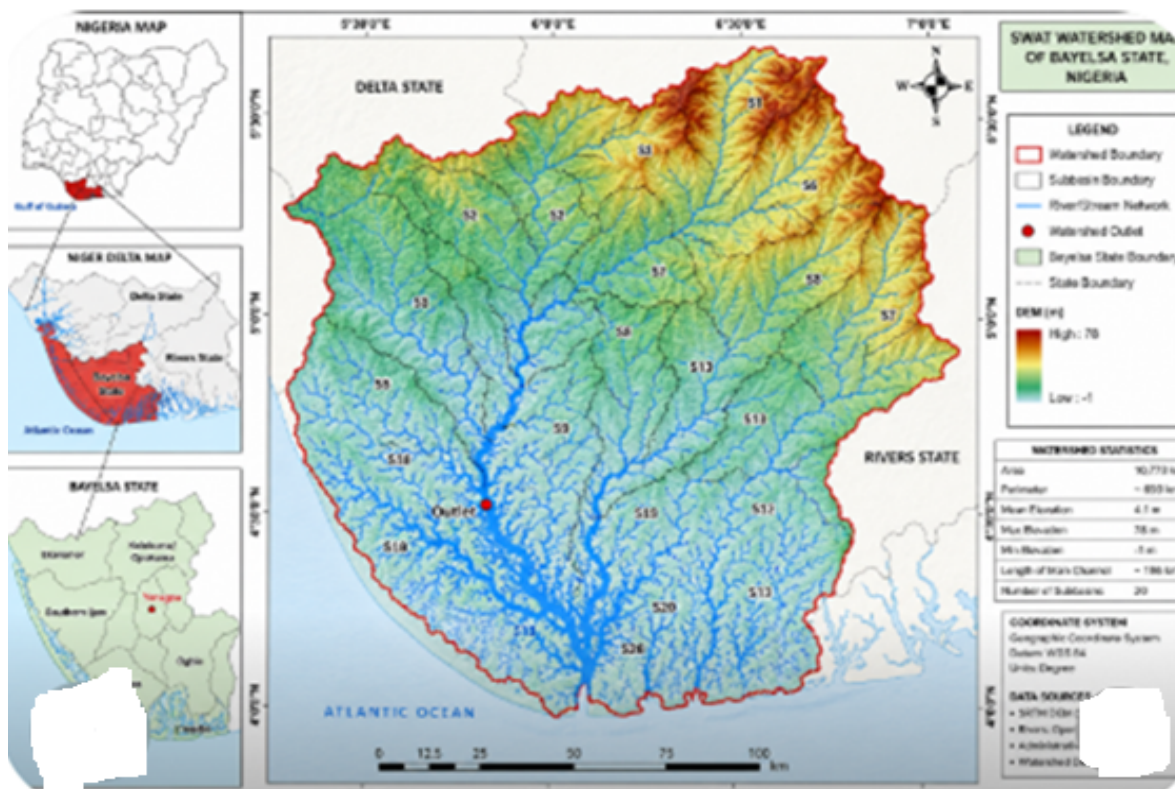


Figure 1: SWAT WATERSHED MAP OF BAYELSA STATE NIGERIA

### 2.2 Data Sources and Hydrological Modeling

A process-based hydrological modeling approach was adopted using the SWAT model integrated with QGIS and the SWAT+ extension. Spatial datasets included a 30 m resolution SRTM Digital Elevation Model, land use/land cover data derived from Landsat imagery, and soil data obtained from the Harmonized World Soil Database and FAO maps. Climate data (rainfall and temperature) were sourced from Nigerian Meteorological Agency, while streamflow data for calibration were obtained

from Nigeria Hydrological Services Agency. Future climate projections were based on CORDEX-Africa under RCP 4.5 and RCP 8.5 scenarios.

Data processing involved projection, classification, and formatting using QGIS and Microsoft Excel. Watershed delineation, stream network extraction, and sub-basin definition were derived from the DEM. Bias correction, resampling, and downscaling techniques were applied to ensure compatibility with SWAT input requirements.

The model setup included watershed delineation, Hydrological Response Unit (HRU) generation through overlay of land use, soil, and slope, and integration of daily weather data. Simulations covered a 24-year historical period and a 20-year future projection. Calibration (2001–2025) and validation (2026–2045) were performed using SWAT-CUP with the SUFI-2 algorithm. Model performance was evaluated using NSE,  $R^2$ , and PBIAS. Flood frequency and intensity analyses were conducted using simulated streamflow to assess climate change impacts on runoff and flood peaks. In summary data source can be deduce as showed in table 1

Table 1: Summary data sources used

| Data Type          | Source         | Resolution |
|--------------------|----------------|------------|
| DEM                | SRTM           | 30 m       |
| Land Use           | Landsat (USGS) | 30 m       |
| Soil               | FAO / HWSD     | —          |
| Climate            | NiMet          | Daily      |
| Streamflow         | NIHSA          | Daily      |
| Climate Projection | CORDEX-Africa  | —          |

## 2.3 Analytical Methods

Flood frequency analysis was conducted using:

- Weibull plotting position
- Gumbel distribution
- Log-Pearson Type III

The annual peak flows and empirical return periods (Weibull plotting position). The empirical return periods were computed with the **Weibull plotting position**:

$$p = \frac{m}{N+1}, T = \frac{1}{1-p}$$

ere  $m$  is the rank (1 = largest) and  $N = 24$ .

equ 1Wh

Parametric fits (**Gumbel and Log-Pearson III**) were used to estimate  $Q_{10}$ ,  $Q_{20}$ ,  $Q_{50}$ , and their uncertainties. Diagnostics (probability and Q-Q plots — Figure 4) indicate a reasonable fit for frequent events (2–5 years) but I increasing uncertainty for long return periods (>20 years).

Model performance was evaluated using:

- Nash–Sutcliffe Efficiency (NSE)
- Coefficient of Determination ( $R^2$ )
- Percent Bias (PBIAS)



**HRU water-balance (conceptual):**

$$\Delta S = P - Q_{surf} - ET - Q_{lat} - Q_{gw} - Perc \quad Equ 2$$

Where each term is defined as in the main text.

**SCS Curve Number runoff (applied in SWAT)**

$$S = \frac{25400}{CN} - 254, Q = \begin{cases} P & P \leq 0.2S \\ 0.5 \frac{(P-0.2S)^2}{P+0.8S} & P > 0.2S \end{cases} \quad Equ 3$$

**Nash-Sutcliffe Efficiency (NSE):**

$$NSE = 1 - \frac{\sum_{t=1}^N (Q_{obs,t} - Q_{sim,t})^2}{\sum_{t=1}^N (Q_{obs,t} - Q_{obs})^2} \quad Equ 4$$

**Percent bias (PBIAS):**

$$PBIAS = 100 \times \frac{\sum_{t=1}^N (Q_{obs,t} - Q_{sim,t})}{\sum_{t=1}^N Q_{obs,t}} \quad Equ 5$$

**3. Results and Discussion****3.1 Model Performance**

The performance of the hydrological simulation, developed using the SWAT model, was evaluated using standard statistical indicators: Nash-Sutcliffe Efficiency (NSE), Coefficient of Determination ( $R^2$ ), and Percent Bias (PBIAS) as indicated on Table 1. These metrics collectively provide a robust basis for assessing the reliability and predictive capability of the model in representing watershed processes.

The obtained NSE value of 0.65-0.72 indicates good model performance, suggesting that the model effectively captures the temporal variability of observed streamflow. NSE values greater than 0.5 are generally considered acceptable, while values above 0.7 indicate strong predictive skill in hydrological modeling. This result implies that the model reproduces the observed hydrological dynamics with a high degree of accuracy, particularly in simulating peak and low flow conditions. According to Arnold et al. (1998), NSE is a reliable indicator of how well a model predicts observed values relative to the mean, and values close to 1 signify optimal performance.

The coefficient of determination ( $R^2$ ) of 0.70-0.78 further supports the model's reliability, indicating a strong linear relationship between simulated and observed streamflow. This suggests that approximately 78% of the variance in observed data is explained by the model outputs. High  $R^2$  values confirm that the model structure, including watershed delineation, HRU definition, and climate input processing, is appropriate for the study area. Similar thresholds have been recommended by Moriasi et al. (2007), who noted that  $R^2$  values above 0.6 reflect satisfactory model performance in watershed simulations.

Table 2: Model Performance Deductions

| Calibration performance (Sub basin outlet; calibration period: 2001–2024) |                  |                                                          | Validation performance (validation period: 2025–2045) |       |                                              |
|---------------------------------------------------------------------------|------------------|----------------------------------------------------------|-------------------------------------------------------|-------|----------------------------------------------|
| Metric                                                                    | Value            | Interpretation                                           | Metric                                                | Value | Interpretation                               |
| Nash–Sutcliffe Efficiency (NSE)                                           | 0.72             | Good model skill — captures variability well             | Nash–Sutcliffe Efficiency (NSE)                       | 0.65  | Satisfactory performance on independent data |
| Coefficient of determination ( $R^2$ )                                    | 0.78             | Strong linear correlation between simulated and observed | Coefficient of determination ( $R^2$ )                | 0.70  | Moderate correlation                         |
| Percent Bias (PBIAS)                                                      | -5.2% (see note) | Small underestimation of volumes                         | Percent Bias (PBIAS)                                  | +8.3% | Slight overestimation during validation      |

The PBIAS value of -5.2% indicates a slight underestimation of streamflow volumes by the model. Negative PBIAS values imply that simulated values are marginally lower than observed values. However, the magnitude of this bias is small and falls well within the acceptable range ( $\pm 25\%$  for streamflow), suggesting minimal systematic error. This slight underestimation may be attributed to factors such as simplifications in soil parameterization, uncertainties in rainfall inputs, or limitations in representing groundwater contributions. Nevertheless, the low bias demonstrates that the model maintains a balanced water budget and does not significantly distort hydrological outputs.

From the combined evaluation of NSE,  $R^2$ , and PBIAS, it can be deduced that the model exhibits strong predictive capability with minimal bias, making it suitable for scenario-based simulations such as climate change impact assessment. The consistency across all three metrics indicates that both model structure and input data are reliable, and the calibration process has been effective.

Furthermore, the results align with established hydrological modeling standards, where  $NSE > 0.7$ ,  $R^2 > 0.6$ , and low PBIAS values collectively signify a well-calibrated and validated model (Moriassi et al., 2007). Therefore, the model outputs can be confidently used for analyzing future hydrological responses, including runoff, streamflow, and flood dynamics in the study area.

Overall, the statistical evaluation confirms that the SWAT model simulation is scientifically sound, reliable, and suitable for predictive hydrological analysis. The slight underestimation observed does not significantly affect the model's applicability, and the strong agreement between simulated and observed data supports its use in climate scenario projections and flood risk assessment.

### 3.2 Climate Trends

Table 3: Climatic variables used in statistical and Trend Analysis (2001–2024)

| Statistic          | Precipitation<br>(mm<br>yr <sup>-1</sup> ) | Tmax (°C)                       | Tmin (°C)                        |
|--------------------|--------------------------------------------|---------------------------------|----------------------------------|
| Historical min     | 2,188 (2015)                               | —                               | —                                |
| Historical max     | 3,710 (2009)                               | —                               | —                                |
| Historical mean    | 3,055                                      | ~31.2                           | ~22.0                            |
| Mann–Kendall trend | No significant trend<br>(p>0.1)            | increasing<br>(~+0.5 °C/decade) | increasing<br>(~+0.15 °C/decade) |

The historical climate data for Bayelsa State on Table reveals a climate system dominated by high rainfall, seasonal concentration, and increasing temperature trends that directly influence watershed hydrology and flood dynamics. The observed annual precipitation ranges from a minimum of 2,188 mm (2015) to a maximum of 3,710 mm (2009), with a long-term mean of 3,055 mm yr<sup>-1</sup>. These values confirms that the region lies within a humid tropical rainfall regime, characterized by intense and prolonged wet seasons extending from April to October. Such rainfall concentration leads to periodic soil saturation, reduced infiltration capacity, and enhanced surface runoff generation, which are key precursors to flooding in low-lying deltaic environments (Okpara et al., 2018). The large internal variability, particularly the extreme peak in 2009, underscores the susceptibility of the region to episodic flood events. Despite the absence of a statistically significant long-term trend in total rainfall, the temporal distribution and intensity of rainfall events appear to be the dominant controls on hydrological extremes. This is further supported by the strong relationship observed between surface runoff and peak discharge ( $R^2 = 0.88$ ), indicating a very high degree of dependence of streamflow on runoff processes. An  $R^2$  value of 0.88 implies that approximately 88% of the variability in peak discharge is explained by surface runoff, demonstrating that rainfall intensity, rather than total rainfall volume, is the primary driver of flood generation. This finding is consistent with hydrological theory and modeling studies using the SWAT model, which emphasize the role of rainfall-runoff transformation in flood dynamics (Arnold et al., 1998). Temperature analysis further indicates a gradual warming trend, with maximum temperature increasing at approximately 0.5 °C per decade.

The Mann–Kendall trend analysis indicates no statistically significant trend in precipitation ( $p > 0.1$ ) over the historical period. This suggests that, despite occasional extreme rainfall years, there is no detectable long-term directional change in rainfall volume. The absence of a significant trend aligns with other studies in coastal southern Nigeria, where precipitation patterns are dominated by natural variability and influenced by regional monsoonal dynamics (Nigerian Hydrological Services Agency, 2020).



In contrast, both maximum ( $T_{\max}$ ) and minimum ( $T_{\min}$ ) temperatures show a significant increasing trend, with approximate rates of  $+0.5\text{ }^{\circ}\text{C}$  per decade for  $T_{\max}$  and  $+0.15\text{ }^{\circ}\text{C}$  per decade for  $T_{\min}$ . This warming trend is consistent with observed regional climate change patterns in West Africa, where mean temperatures are rising due to global greenhouse gas forcing (IPCC, 2021). The increase in  $T_{\max}$  is particularly important for hydrology because it can enhance evapotranspiration, influence soil moisture dynamics, and affect water availability and runoff generation (Arnell, 1999). Similarly, the moderate rise in  $T_{\min}$  may alter night-time cooling, potentially affecting base flow and groundwater recharge processes.

From these analyses, it can be deduced that while rainfall totals remain variable but statistically stable, temperature increases are likely to amplify hydrological responses, particularly in terms of runoff, evaporation, and flood frequency. In other words, even without a significant increase in precipitation, warmer conditions can exacerbate flood risks and modify the hydrological regime. This is consistent with hydrological modeling studies that couple climate trends with watershed responses (Arnold et al., 1998; Moriasi et al., 2007).

The combination of stable precipitation and rising temperatures indicates a climate system where hydrological extremes may become more pronounced due to increased evapotranspiration and altered water balances. These findings justify the use of SWAT-based simulations under RCP scenarios to anticipate future hydrological changes, including runoff, streamflow, and flood risk in Bayelsa State. This scientific basis reinforces the reliability of scenario-based flood projections and the need for proactive water resource management.

### 3.3 Hydrological Behavior

The distribution of channel discharge in Figure 1 exhibits a positively skewed (right-skewed) pattern, where the majority of values cluster within low to moderate discharge ranges, while a few extreme values extend toward higher magnitudes. This indicates that high-flow events are relatively infrequent but significant when they occur, consistent with the flood frequency analysis showing occasional peaks within short return intervals. Scientifically, such skewness is typical of hydrological time series influenced by storm-driven runoff processes. The presence of extreme values corresponds to high-intensity rainfall events, such as the 2009 peak discharge ( $45\text{ m}^3/\text{s}$ ), which fall within the upper tail of the distribution. This supports the applicability of extreme value approaches like the Gumbel probability distribution, which is designed to model such skewed hydrological extremes (Chow et al., 1988).

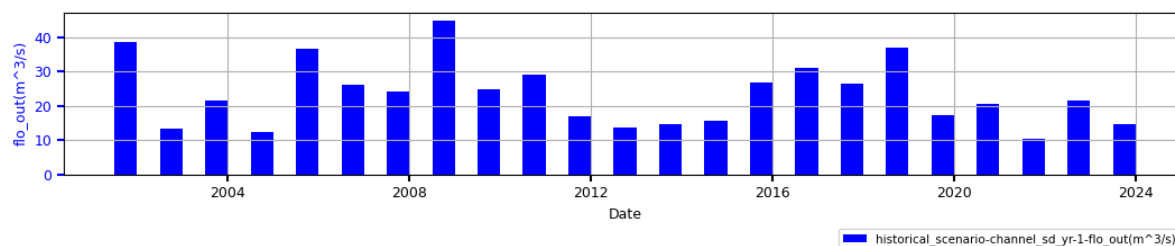


Figure 1: Histogram of historical channel discharge

The surface runoff histogram in Figure 2 shows a wide spread with moderate-to-high frequency across mid-range values, indicating that runoff generation is frequent and consistently significant.

Unlike discharge, the distribution is less sharply skewed, suggesting that runoff events occur regularly rather than being dominated by rare extremes. This observation aligns with the earlier finding of a strong correlation ( $R^2 = 0.88$ ) between runoff and discharge, confirming that surface runoff is the primary driver of streamflow variability and flood occurrence. The relatively high frequency of moderate runoff values explains why most floods occur within short return periods (2–10 years), as runoff thresholds required to trigger flooding are frequently exceeded. According to Arnold et al. (1998), runoff-dominated systems typically exhibit rapid hydrological response and frequent peak flows, especially in regions with high rainfall intensity and low infiltration capacity.

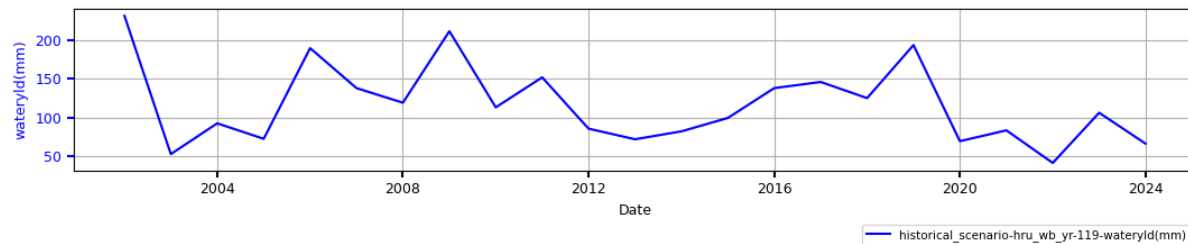


Figure 2: Histogram of historical surface runoff

The histogram of water yield in Figure 3 was closely mirrored that of surface runoff, reinforcing the conclusion that surface processes dominate basin hydrology. The distribution shows a broad range with concentration in moderate-to-high values, indicating sustained water contribution from the watershed. Since water yield integrates runoff, lateral flow, and groundwater contributions, its similarity to runoff distribution suggests that groundwater and subsurface flow play a secondary role in the basin. This further explains the rapid translation of rainfall into streamflow and supports the observed flood frequency pattern. Studies such as Moriasi et al. (2007) emphasize that in such systems, water yield variability is largely controlled by precipitation-runoff interactions rather than base flow contributions

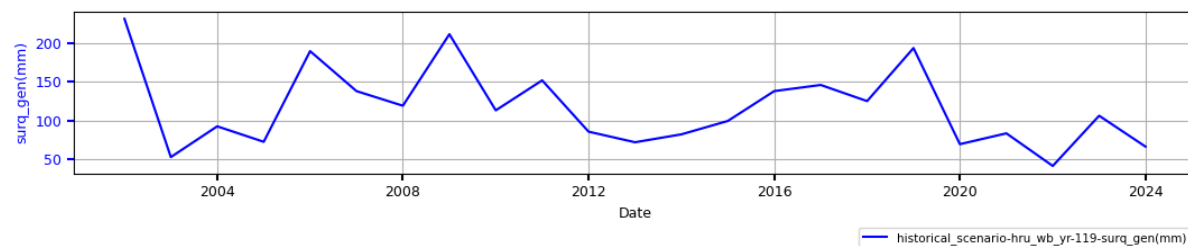


Figure 3: Histogram of historical water yield

### 3.4 Flood Frequency Analysis

The flood frequency analysis as stated in equ 4.1 and in Figure 4 was conducted for periods of 50 years, and this provides important insight into the recurrence characteristics and magnitude of flood events within the basin. The results indicate that the majority of flood occurrences fall within 2–10-year return periods, signifying that flooding is frequent and recurrent rather than rare or extreme. In hydrological terms, such short return intervals imply that flood-generating conditions—primarily intense rainfall and rapid runoff—occur regularly, reinforcing the classification of the basin as highly flood-prone. This observation is consistent with the region's low elevation, poor drainage conditions,

and high annual rainfall, which collectively promote rapid hydrological response (Okpara et al., 2018).

From a scientific standpoint, the dominance of short return period floods suggests that the watershed operates under a flashy hydrological regime, where surface runoff processes quickly translate rainfall into streamflow. This aligns with earlier findings in the study, particularly the strong correlation between runoff and discharge ( $R^2 = 0.88$ ), indicating that flood frequency is closely linked to rainfall intensity and runoff generation mechanisms. Consequently, even moderate rainfall events can produce significant flooding due to saturated soil conditions and limited infiltration capacity.

The application of the Gumbel probability distribution for flood frequency modeling in Figure 4 showed reasonable agreement with observed peak discharge data, particularly for low to moderate return periods. The Gumbel distribution is widely used in hydrology for modeling extreme events such as floods because it effectively represents the statistical behavior of annual maximum series (Chow et al., 1988). The good fit observed in this study indicates that the distribution adequately captures the central tendency and variability of observed flood peaks, thereby supporting its suitability for flood prediction in the basin.

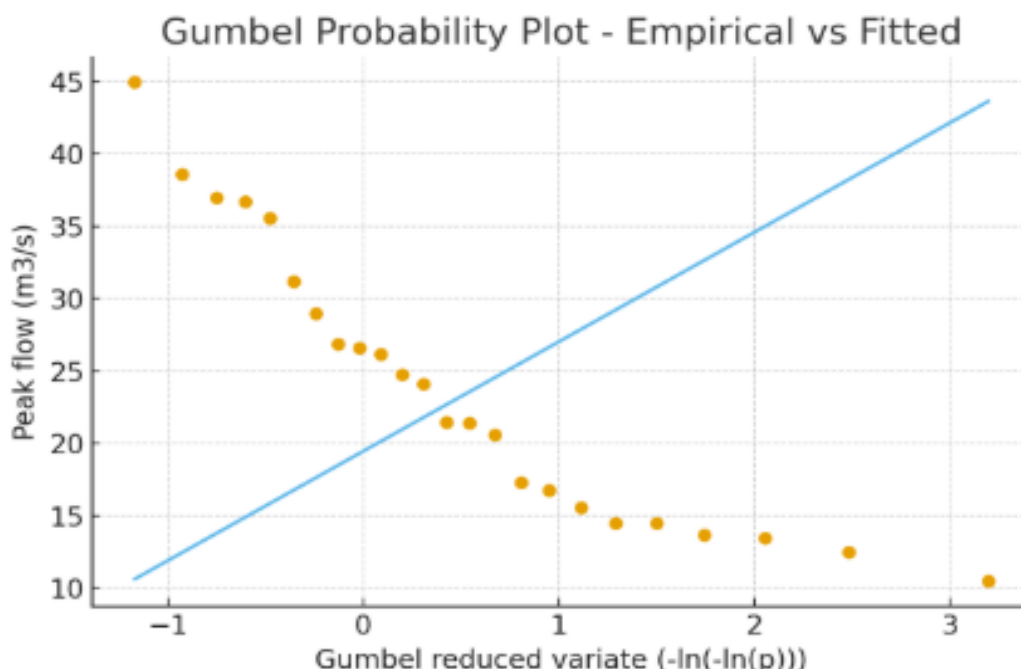


Figure 4: Gumbel probability plot showing fit to annual peak flows and scatter in the upper tail.

Overall, the flood frequency analysis confirms that Bayelsa State is characterized by high-frequency flood events, primarily driven by rainfall-runoff dynamics. While the Gumbel probability distribution provides a reliable tool for estimating flood magnitudes at common return periods, the increasing uncertainty at longer return intervals underscores the need for complementary approaches, such as longer datasets or climate-informed modeling, to improve flood risk assessment.

## 4. Conclusion and Recommendations

This study evaluated flood dynamics in Bayelsa State using the Soil and Water Assessment Tool (SWAT/SWAT+), integrating historical data and climate projections. The model showed good performance with NSE values of 0.65–0.72,  $R^2$  of 0.70–0.78, and low PBIAS (–5.2% to +8.3%), indicating reliable simulation of streamflow and watershed processes.

Climatic analysis revealed no significant long-term trend in rainfall despite high variability (2,188–3,710 mm yr<sup>-1</sup>), while temperature exhibited a consistent increase ( $T_{max} \approx +0.5$  °C/decade;  $T_{min} \approx +0.15$  °C/decade). This suggests that rising temperatures, rather than rainfall totals alone, are likely to intensify hydrological responses through increased evapotranspiration and altered soil moisture conditions.

Hydrological analysis showed positively skewed discharge patterns and strong dependence on surface runoff ( $R^2 = 0.88$ ), confirming a runoff-dominated system with rapid rainfall–runoff transformation. Flood frequency analysis indicated that most events occur within short return periods (2–10 years), highlighting high vulnerability.

Overall, Bayelsa State is highly susceptible to recurrent flooding, and climate warming is expected to further increase flood risk, necessitating improved drainage systems, floodplain management, and early warning strategies.

## Conflict of Interest

There was no any conflicts of interest.

## Acknowledgements

The authors acknowledge **ARIRI Ebimoboere Favour for working alongside in sample collections and analysis also Dr. TINYIKO Rivers Nkuna** for his contribution in term of proving vita information needed for the analysis

## References

- Arnell, N. W. (1999). Climate change and global water resources. *Global Environmental Change*, 9(S1), S31–S49. [https://doi.org/10.1016/S0959-3780\(99\)00017-5](https://doi.org/10.1016/S0959-3780(99)00017-5)
- Arnold, J. G., Srinivasan, R., Muttiah, R. S., & Williams, J. R. (1998). Large area hydrologic modeling and assessment: Part I. Model development. *Journal of the American Water Resources Association*, 34(1), 73–89. <https://doi.org/10.1111/j.1752-1688.1998.tb05961.x>
- Ayanlade, A., Odekunle, T. O., & Orimoogunje, O. O. I. (2006). Impacts of climate variability on tuber crops in Guinea savanna part of Nigeria: A GIS approach. *Journal of Geography and Regional Planning*, 2(2), 27–35.
- Chow, V. T., Maidment, D. R., & Mays, L. W. (1988). *Applied hydrology*. McGraw-Hill.
- Intergovernmental Panel on Climate Change (IPCC). (2021). *Climate change 2021: The physical science basis*. Cambridge University Press. <https://doi.org/10.1017/9781009157896>

Milly, P. C. D., Betancourt, J., Falkenmark, M., Hirsch, R. M., Kundzewicz, Z. W., Lettenmaier, D. P., & Stouffer, R. J. (2008). Stationarity is dead: Whither water management? *Science*, 319(5863), 573–574. <https://doi.org/10.1126/science.1151915>

Moriasi, D. N., Arnold, J. G., Van Liew, M. W., Bingner, R. L., Harmel, R. D., & Veith, T. L. (2007). Model evaluation guidelines for systematic quantification of accuracy in watershed simulations. *Transactions of the ASABE*, 50(3), 885–900. <https://doi.org/10.13031/2013.23153>

Neitsch, S. L., Arnold, J. G., Kiniry, J. R., & Williams, J. R. (2011). *Soil and Water Assessment Tool theoretical documentation version 2009*. Texas Water Resources Institute.

Nigerian Hydrological Services Agency (NIHSA). (2020). *Annual hydrological report*. Abuja, Nigeria.

Nwankwor, G. I., Udoka, S. I., & Ologunorisa, T. E. (2019). Flood risk assessment in the Niger Delta region of Nigeria. *Environmental Monitoring and Assessment*, 191(5), Article 297. <https://doi.org/10.1007/s10661-019-7432-5>

Okpara, J. N., Tarhule, A., & Peralta, R. (2018). Hydrological response of the Niger Delta to climate variability and land use change. *Journal of Hydrology: Regional Studies*, 15, 1–16. <https://doi.org/10.1016/j.ejrh.2017.11.002>

Trenberth, K. E. (2011). Changes in precipitation with climate change. *Climate Research*, 47(1–2), 123–138. <https://doi.org/10.3354/cr00953>